

ASHWIN K K

Electronics and communication engineer

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SUMMARY

Electronics and communication engineer with hands-on knowledge in the areas of embedded systems, IoT, and wireless communication systems. Proficient in Embedded C, ESP32, Arduino, PCB schematic and layout, soldering and testing of electronic components with a solid understanding of communication protocols. seeking opportunities to contribute to innovative and reliable engineering solutions.

SKILLS

- **Programming Languages:** Embedded C, SQL
 - **Communication Protocols:** UART, SPI, I2C
 - **Tools:** kiCAD, MPLAB, MS office
 - **Soft skills:** Communication, Team work, Observation, Creativity, Adaptability
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EDUCATION

B. Tech in Electronics and Communication Engineering College of Engineering Thalassery	2022 – 2026
Diploma in electronics engineering Govt. polytechnic collage mattanur	2020 - 2022

PROJECTS

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|--|-------------|
| LORA based V2X communication system | 2025 - 2026 |
| <ul style="list-style-type: none">• Developed a V2X communication prototype implementing V2V, V2I, and V2P scenarios using ESP32, Arduino Nano, and LoRa (SX1278), enabling collision warnings, traffic-signal status updates, and pedestrian/ambulance alerts. | |
| Blackbox system for emergency response | 2024 - 2025 |
| <ul style="list-style-type: none">• developed a vehicle SOS prototype system called Blackbox that can implement on trucks, cars etc. to respond if an accident occur.• provide alert message to the system to the emergency contacts including exact date and time with location details. also store the accident details with timestamp in a memory.• Observe the accuracy of the sensors by changing the sensor values and detect the response time. | |
| Vision based intelligent automation system | 2021 - 2022 |
| <ul style="list-style-type: none">• Developed a vision-based automation system using yolo algorithm to detect people faces.• Automatically control the fan or light in any office, classroom and other halls.• Save the energy and also count the peoples inside the hall with better accuracy. | |
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CERTIFICATES

Technical Support Fundamentals [Google]

- clear understanding of the technical support in IT field by approach and analyzing the problems, develop solutions based on real world scenarios and confident in solving technical issues.

Introduction to IoT and embedded systems [University of California, Irvine]

- Knowledge of network of physical devices and other technologies to connect and exchange data between connected objects. learned about the hardware and software like processors memories, sensors various communication protocols.

Embedded system Design and Programming [Invade embedded technologies]

- Done by an internship program at invade embedded technologies.
- Worked in interfacing of different electronics components, programming in PIC microcontroller, and PCB design.